

DevOps Take-Home Assignment

Objective: Design and deploy a Task Tracker API service using DevOps best practices - containerized, infrastructure-managed, automated through CI/CD, and monitored.

Project Scope:

1. Application Setup

- Build a Task Tracker API (Flask, FastAPI, or Node.js).
- Required endpoints:
- POST /tasks - Create a task
- GET /tasks - List all tasks
- Use SQLite or PostgreSQL for persistence.

2. Infrastructure Provisioning (Terraform)

- Provision the following on AWS:
- 1 x EC2 instance (Ubuntu 22+)
- 1 x Security Group (allowing port 80/3000)
- 1 x S3 bucket (optional: logs or artifacts)
- Output public IP and bucket name.

3. App Deployment

- Dockerize the app
- Use Ansible (or Shell scripts) to:
- Install Docker on EC
- Pull container image
- Run it as a service

3. CI/CD Pipeline

- Use GitHub Actions or GitLab CI/CD to:
- Run tests on push to main
- Build Docker image
- Push to Docker Hub or GHCR
- SSH into EC2 and deploy

4. Monitoring

- Install Prometheus Node Exporter or Grafana Agent
- Optional: Logs to S3 or app-level /metrics endpoint
- Bonus: Grafana dashboard

Deliverables:

- GitHub/GitLab repository with:
- Terraform scripts
- Dockerfile
- Deployment scripts (Ansible/Shell)
- CI/CD config
- README with setup steps and diagram

README Requirements:

- Architecture diagram (image or ASCII)
- Setup steps for:
- Infra
- App deployment
- API testing
- Monitoring
- Notes on design decisions or challenges

Evaluation Criteria:

- Code and documentation clarity
- CI/CD completeness
- Automation and infra provisioning
- Monitoring integration

Bonus points for:

- Zero-downtime deploys
- Use of ECS/Fargate
- Self-healing infra

